
PROFESSIONAL SUMMARY

Software Engineer with 5+ years delivering production-grade **distributed systems and microservices architectures** across enterprise platforms. Deep expertise in full-stack development, REST API design, and cloud infrastructure (**AWS, Kubernetes, Docker**) with proven ownership of end-to-end system design, **scalability optimization**, and cross-functional technical leadership. Strong foundation in modern backend technologies (**Node.js, Java/SpringBoot, Python**), database optimization, and building **high-availability systems** driving operational efficiency and business impact at scale.

PROFESSIONAL EXPERIENCE

Full Stack Software Engineer

Hub Group Inc | Jan 2024 - Present

- Led design and implementation (**with 1 senior engineer and 2 SDEs**) of **microservices-based** billing platform processing **\$2B+ annual invoice volume**, automating invoice processing and **ERP integration** while retiring legacy **COBOL** systems.
- Architected migration (**over 10 months, across 12 microservices**) from **monolithic COBOL** to containerized Node.js microservices on Kubernetes/Docker, reducing costs by **\$500K** annually while enabling independent scaling and **zero-downtime deployments**.
- Engineered **REST API** integration between Order Management platform and TMS using custom **ETL pipelines** processing **50K+ daily orders**, eliminating manual data transfers and driving an **estimated \$1M revenue growth** through operational efficiency.
- Designed and built real-time order failure detection system using **Azure Service Bus topics** and automated recovery workflows, improving order completion rate to 99.8% and reducing manual intervention by **85%** across **100K+ monthly transactions**.
- Redesigned **distributed API architecture** to support **10x traffic growth**, implementing **Redis clustering** and multi-tier **caching strategies** to handle **500K+ monthly requests**, reducing P95 response time by **60% (800ms → 320ms)**.
- Built anomaly detection system using threshold-based alerting and ML-augmented baselines (**TensorFlow**), integrated with **Elasticsearch/Kibana** dashboards to detect **15+ critical incidents**, reducing MTTD to **under 2 minutes**.
- Optimized **Node.js/Express microservices** and **MongoDB** aggregation pipelines through database indexing and query refactoring, improving **API response times by 40%** and reducing P99 latency (**1.2s → 720ms**) for **50K daily users**.
- Automated **zero-downtime deployment** pipeline using **Kubernetes rolling updates via Jenkins**, coordinating releases across 8 microservices with automated health checks, and rollback mechanisms, **reducing release cycle (4 hours → 1 hour)**.

Software Engineer

Cerner Corporation [Oracle Health] | Apr 2021 - Aug 2022

- Implemented **HL7 v2.x** and **FHIR** message handlers processing **150K+ daily patient transactions**, reducing interface transmission errors by **22%** through enhanced validation logic and error handling across production **EHR integrations**.
- Created custom **Discern Rules** and **SQL-based** reporting solutions against **Oracle Database** for revenue cycle and scheduling workflows, delivering **15 production reports** serving **500+ clinical and operational staff** members daily.
- Designed an automated clinical reporting framework using **CCL** and **Oracle Database**, deployed across **8 hospital clients** serving **10K+ clinicians**, reducing manual data reconciliation by **45%** (saving **20 staff-hours/week** per site).

Programmer Analyst

Cognizant | Sep 2020 - Apr 2021

- Built full-stack financial platform using **React.js** and **Spring Boot microservices** with **OAuth2/JWT** authentication and **role-based access control (RBAC)**, serving **10K+ daily active users** while maintaining **SOC2 compliance** and **99.9% uptime**.
- Engineered high-throughput **RESTful APIs** using **Spring Security** with **Redis caching** and custom interceptors, reducing API latency by **35%** and implementing **rate limiting** for critical World Bank integration endpoints processing **50K+ daily requests**.
- Architected hybrid database solution using **MySQL sharding strategies** and **indexing**, improving query performance by **40%** while ensuring ACID compliances, supporting **100M+ records** across distributed financial transaction systems.

TECHNICAL SKILLS

Programming Languages: JavaScript, Python, TypeScript, Java, SQL, CCL, C++**Frontend Technologies:** React.js, Vue.js, Next.js, Angular, HTML5, CSS3, State Management (Redux, Vuex)**Backend & APIs:** Node.js, Express, Spring Boot, REST APIs, GraphQL, API Design, WebSockets**System Design & Architecture:** Distributed Systems, Microservices Architecture, Event-Driven Architecture, Service-Oriented Architecture (SOA), System Scalability, High Availability, Load Balancing, Service Mesh, API Gateway, Fault Tolerance**Cloud & DevOps:** Microsoft Azure, AWS (EC2, Lambda, S3, RDS, DynamoDB, API Gateway, CloudWatch), Kubernetes, Docker, CI/CD Pipelines, Jenkins, GitHub Actions, Infrastructure as Code (Terraform), Container Orchestration**Databases & Data Storage:** MongoDB, PostgreSQL, MySQL, Oracle Database, SQL Server, Redis (Caching)**Message Queues & Streaming:** Service Bus, Apache Kafka, RabbitMQ, Event Streaming, Asynchronous Processing, ETL Pipelines**Security & Authentication:** OAuth2, JWT, Role-Based Access Control, API Security, Data Encryption, HIPAA/SOC2 Compliance**Observability & Reliability:** Elasticsearch, Kibana, Prometheus, Logging (ELK Stack), Metrics & Monitoring, Alerting, Distributed Tracing, Performance Optimization, Application Reliability Engineering, Incident Response, On-Call Rotation**Testing & Quality:** Unit Testing, Integration Testing, Test-Driven Development (TDD), SonarQube, Automated Testing**AI & Machine Learning:** FastAPI, TensorFlow, LLM Workflows, GenAI Application Integration, Agent Orchestration**Tools & Practices:** Git, Agile/Scrum, RESTful Design Patterns, Code Review, Technical Documentation, System Troubleshooting

EDUCATION

Master of Computer Science - Illinois Institute of Technology, USA